HK IMO Training 2022-2023
Problem Set 4
For 15 October 2022

Problem 13. We colour all positive integers such that the following three conditions hold:

- all the odd positive integers are red;
- n has the same colour as $4n$ for every positive integer n;
- n has the same colour as $n+2$ or $n+4$ for every positive integer n.

Prove that all positive integers are red.

Problem 14. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xy) = \max\{f(x+y), f(x)f(y)\}$$

for all real numbers x and y.

Problem 15. Let $\triangle ABC$ be a non-equilateral triangle with orthocentre H and circumcentre O. Let ω_1 be the circle passing through B and tangent to AC at A. Let ω_2 be the circle with centre lying on the line BH and tangent to AB at A. The circles ω_1 and ω_2 meet again at P. Prove that $\angle HPO = 180^\circ - \angle BAC$.

Problem 16. Let α and β be two irrational numbers. Let A be the set of all positive integers n such that $\left|\alpha - \frac{m}{n}\right| < \frac{1}{10n}$ for some integer m. Let B be the set of all positive integers n such that $\left|\beta - \frac{m}{n}\right| < \frac{1}{10n}$ for some integer m. If $A = B$, show that $\alpha + \beta$ or $\alpha - \beta$ is an integer.